MIL-P-15418C(SHIPS)
27 October 1961
SUPERSEDING
MIL-P-15418B(SHIPS)
13 June 1957

MILITARY SPECIFICATION

PLOTTING BOARDS, TACTICAL DISPLAY (SYNCHRO, 24 INCH)

1. SCOPE

1.1 This specification covers 24-inch synchro summary plotting boards for use in combat information centers or similar spaces in Naval ships.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-983 - Interior Communication Equipment; Naval Shipboard, Basic Design Requirements for

MIL-P-15137 - Provisioning and Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).

MIL-E-17362 - Electronic Repair Parts Requirements, Procedures for Stock Numbering and Assignment of Government Nomenclature.

MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of

MIL-S-20708 - Synchros, 60 and 400 Cycle; General Specification.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MS 15562 - Lamps, Incandescent, T-8, Bayonet Candelabra, Single Contact.

MS 15574 - Lamp, Incandescent, T-7, Candelabra Screw.

DRAWINGS

BUREAU OF SHIPS

S2407-566927 - 24 Inch Glass Plotting Board, MK2 MOD 3.

S2407-740804 - Summary Plotting Board, 24 Inch, Synchro MK2 MOD3 - Wiring Diagram

S2407-740805 - Summary Plotting Board, 24 Inch, Synchro MK2 MOD3 - Complete Assembly.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules and Regulations.

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York).

3. REQUIREMENTS

3.1 Preproduction sample - One preproduction sample summary plotting board shall be submitted for the tests specified in 4.2.

3.2 General requirements. - The summary plotting boards shall be in accordance with the following paragraphs of Specification MIL-I-983 in addition to the requirements specified herein. Whenever a requirement of Specification MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern.

General information
Materials, general
Stressed or vital parts
Fungus inert materials
Unacceptable materials
Acceptable materials
Flammable materials
Toxic materials
Wood
Aluminum
Magnesium
Iron and steel
Plastics

FSC 6605

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Lubricants and lubrication
Painting
Protection against corrosion
Dials and pointers
Dial size
Locking devices
Washers
Ball bearings
Parts - electrical
Variable resistors
Transformers
Synchros (see 3.7)
Dial illumination lamps
Enclosures - general
Enclosure - accessibility
Enclosure - degree of
Stiffening grooves
Minimum sheet metal thickness
Through bolting
Stuffing tubes
Ventilation
Size
Threaded devices
Rounded corners and edges
Drilled and tapped holes
Welding
Temperature and humidity
Shock, vibration, and inclination (Shock and vibration only) (nonvital vibration test required)
Primary power circuits
Power supply tolerances (see 3.8)
Personnel protection
Shielding and radio frequency noise reduction
Ground potential and grounding
Soldering
Terminal boards, connectors, and terminals
Wiring
Dielectric strength and insulation resistance clearances
Drawings (general, preliminary, working, and schematic)
List of repair parts
Interchangeability and standardization
Technical manuals
Repair parts
Designation and marking
Design for production
Workmanship and general examination

3.3 Weight - Weight shall be kept to a minimum The maximum weight of the equipment shall not exceed 75 pounds.

3.4 Painting. - Painting shall be in accordance with Specification MIL-I-983, except that the interior finish coat shall be of inside white, semi-gloss, fire retardant paint, in lieu of the gray equipment enamel.

3.5 Design features. - The summary plotting board shall consist of a synchro motor, transformer, rheostats, interior lights, necessary cable, terminals and connection block, and a projection system containing a reticle, lens, mirrors, reflectors, shields and projection lamp, assembled in a drip-proof case fitted with a glass plotting top. The assembly shall be in accordance with Drawing S2407-740805 and the internal wiring shall be in accordance with Drawing S2407-740804.

3.6 Projection system. - The own ship's course projection system in the summary plotting board shall function essentially as follows:

3.6.1 A projector lamp shall illuminate a reticle by reflections from a mirror set an an angle of 45 degrees.

3.6.2 A reticle (see 3.13) shall be mounted on the end of the synchro motor shaft The synchro motor shall receive its operating voltage, and, consequently, indication of ship's heading, from the gyro compass system.

3.6.3 A projection lens shall pick up the image of the reticle through a hole provided in the mirror (see 3 6.1) and shall project the image on a second mirror which in turn reflects it onto the underside of the center circle of the glass plotting top.

3.6.4 Upon a change of ship's heading, the synchro signal from the gyro compass system shall actuate the synchro motor in the summary plotting board by causing a corresponding angular rotation of the shaft and the reticle mounted thereon. By means of the projection system heretofore specified, this change in ship's heading shall be indicated on the plotting top by a corresponding angular rotation of the reflected image of the ship's outline

3.7 Synchros - A synchro motor, type 18TR-6 (60-cycle) or type 18TR-4 (400-cycle) in accordance with Specification MIL-S-20708, as specified (see 6.2), shall be provided to receive the single speed synchro input from the gyro compass system.

3.8 Transformer. - A 115-volt/10-volt, single phase, 60-cycle transformer shall be provided for the purpose of energizing the 10-volt 5-amp projection lamp specified in 3.11.

3.9 Rheostats. - Two variable rheostats, 115-volt, 100-watt, mounted on the front panel of the summary plotting board shall be provided One rheostat shall be used for controlling the intensity of illumination of the projection lamp. The second rheostat shall be used for controlling the intensity of the four general illumination lamps. Rheostats shall be so arranged that the circuit they control shall be turned off when the rheostat knob is rotated to its extreme counterclockwise position.

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3.10 General illumination. - General illumination for the board shall be provided by four lamps in accordance with MS15574-1 connected in parallel. Necessary sockets and reflectors shall be installed.

3.11 Projection lamp. - A projection lamp, in accordance with MS15562-1 together with necessary socket and lamp housing, shall be installed. The lamp housing shall be provided with a ventilating cap for heat dissipation.

3.12 Mirrors. - Two mirrors provided with supports suitable for 45-degree mountings and necessary adjustment devices shall be furnished and installed in accordance with Drawing S2407-740805. Mirrors (first surface type) shall produce clear image reflections as required to accomplish the purpose of the projection system.

3.13 Reticle. - A reticle, with a compass rose and ship's outline engraved thereon, shall be mounted on the end of the synchro motor shaft in such a manner as to permit adjustment by means of set screws in a reticle holder. The ship's outline shall have the longitudinal axis coincident with the 0 to 180-degree line of the compass rose, and the bow shall be pointed towards the zero degree mark.

3.14 Projection lens. - The projection lens shall be of first class quality, and shall produce a clear image projection on the large mirror as required to accomplish the purpose of the projection system. The lens shall be a precision optical part.

3.15 Plotting top. - A glass plotting top 24 by 24 inches shall be provided for this summary plotting board. The top shall be Herculite glass, or equivalent, with a ceramic pencil finish on the top surface only. The top shall be inscribed with polar co-ordinate markings, concentric range circles from 4 to 24 inches, and bearings lines every 10 degrees in accordance with Drawing S2407-566927. The glass plotting surface shall be secured in a metal frame for mounting on the summary plotting board. The plotting surface finish shall be suitable for plotting thereon with a lead pencil of medium hardness, and shall be such as to permit easy erasure of marking. When specified, (see 6.2) a plastic plotting top with a 19-inch rotatable center with similar polar co-ordinate markings shall be provided in lieu of the glass plotting top.

3.16 Accuracy. - The alignment of the image projected by the projection system (see 3.6) with the polar coordinate markings on the plotting top (see 3.15) shall be such that the zero degree marks shall register within plus or minus 1/32 inch when tested in accordance with 4.4.2 1. The error of a transmitted bearing and an indicated bearing shall not exceed 1-1/2 degrees for any one bearing, when tested in accordance with 4.4.2.2. The mean error of all bearings shall not exceed 3/4 of a degree.

3.17 Drawings. - The drawings (and features applicable to each) shall consist of a schematic diagram, a wiring diagram, and assembly drawings.

3.17.1 Appropriate notes shall be added to the drawings giving the following information:

(a) Power requirements under "full load" and "standby" conditions.
(b) Heat dissipation under "full load" and "standby" conditions.

3.17.2 Wiring diagrams. - Wiring diagrams shall include the following information:

(a) All parts in correct relation as to physical location, with terminals clearly shown and marked, and all wiring between parts
(b) The coding of all wires.
(c) Appropriate notes shall designate wire size and type.
(d) Designation of parts shall correspond to those assigned in the diagrams.
(e) Voltage and resistance readings at key service points.

3.17 3 Drawings for provisioning - Drawings required solely for identification and cataloging purposes shall be submitted in accordance with Specification MIL-P-15137 or MIL-E-17362.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection. - Preproduction inspection shall be conducted at a laboratory satisfactory to the bureau or agency concerned. The sample equipment shall be subjected to the following examination and tests which shall be performed in general in the order listed:

Inspection	Requirement paragraph	Test paragraph
General examination	3.2	4.4.1
Supply line voltage and frequency	3.2	MIL-I-983
Radio frequency noise interference	3.2	MIL-I-983

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Inspection	Requirement paragraph	Test paragraph
Temperature and humidity	3.2	MIL-I-983
Accuracy	3.16	4.4.2
Dielectric strength	3.2	MIL-I-983
Insulation resistance	3.2	MIL-I-983
Enclosure	3.2	MIL-I-983
Vibration	3.2	MIL-I-983
Shock	3.2	MIL-I-983

4.2.1 Acceptance - preproduction inspection. - The result of each test shall be compared with the specification requirements. In the event of failure to conform to this specification, the contractor shall correct the cause of failure in the preproduction sample and on future production units.

4.3 Acceptance inspection. - Sample equipment shall be selected in accordance with 4 3.2, 4.3.3, and 4.3.4, and each of the selected equipments shall be subjected to the examination and tests as follows, which shall be performed, in general, in the order listed

Inspection	Requirement paragraph	Test paragraph
Group A examination and tests		
General examination	3.2	4.4.1
Dielectric strength	3.2	MIL-I-983
Insulation resistance	3.2	MIL-I-983
Accuracy	3.16	4.4.2
Group B tests		
Supply line voltage and frequency variation	3 2	MIL-I-983
Radio frequency noise interference	3.2	MIL-I-983
Group C tests		
Temperature and humidity	3.2	MIL-I-983
Enclosure	3.2	MIL-I-983
Vibration	3.2	MIL-I-983
Shock	3.2	MIL-I-983

4.3.1 Inspection lot. - All equipment offered for delivery at one time shall be considered a lot. The lot may include the entire contract quantity or it may be the production of any convenient time period.

4 3.2 Sampling for group A examination and tests. - A sample of equipment shall be selected from each inspection lot in accordance with table I, and shall be subjected to the group A examination and tests specified herein, with lot acceptance based on the sampling requirements of Standard MIL-STD-105.

4.3.3 Sampling for group B tests. - A sample of equipments shall be selected from each inspection lot in accordance with table II, and shall be subjected to each of the group B tests specified herein, with lot acceptance based on the sampling requirements of Standard MIL-STD-105.

4.3.4 Sampling for group C tests. - Group C tests will be required by the bureau or agency concerned when the basic design of the equipment or the material of a vital part has been changed. One complete equipment shall be subjected to each of the group C tests listed herein with acceptance based on 4 3.6

4.3.5 Acceptance - groups A and B examination and tests. - The results of each examination and test shall be compared with specification requirements. Failure to conform to the specification requirements shall be counted as a defect, and the equipment shall be rejected. If the number of such nonconforming equipments exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected

4.3.6 Acceptance - group C tests. - The results of each test shall be compared with specification requirements. In the event of failure to conform to this specification for any group C test, the contractor shall correct the cause of failure on future production units and repair the deficiency in all equipments produced on the contract or order.

4.4 Test procedures. -

4.4.1 General examination. - The general examination shall be performed in accordance with Specification MIL-I-983. The operating test shall be performed as follows The plotting board under test shall be energized from proper sources of electrical supply to the projection and internal lighting circuits and the synchro energized by a type 5G-60 cycle (or an 18CX-400 cycle) synchro test transmitter. General illumination and projection lamps and rheostat controls shall be checked for proper operation. The test transmitter shall be rotated, and the synchro in the equipment under test shall be checked to insure that the instrument synchronizes or follows, and that the direction of rotation is correct.

4.4.2 Accuracy test. -

4.4.2.1 The test transmitter shall be set on zero. Under this condition the zero degree graduation on the projected image shall be lined up with

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the zero degree mark on the plotting surface as required in 3.16. Due to inherent inaccuracies in the plotting glass it may be that the 180-degree markings will not match up under the foregoing condition. In that event the zero setting should be adjusted to one side or the other slightly in order to average out the observed inaccuracy at 180 degrees. In some cases, in order to average this error, it may be necessary and is permissible to exceed the allowable 1/32-inch error by a small amount.

Table I - Sampling for group A examination and tests
AQL (approx.) 1 5 percent defective.

Number of equipments in inspection lot	Number of equipments in sample	(Number of equipments nonconforming on any group A examination or test	
		Acceptance number	Rejection number
7 and under	all	-	-
8 to 15	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 and over	50	2	3

Table II - Sampling for group B tests.

Number of equipments in inspection lot	Number of equipments in sample	(Number of equipments nonconforming on any group B test)	
		Acceptance number	Rejection number
3 and under	all	-	-
4 to 15	3	0	1
16 to 40	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 to 500	15	1	2
501 and over	25	2	3

4.4.2.2 By means of the test transmitter, the projected image in the equipment under test shall be rotated through 360 degrees at 10-degree intervals clockwise and counterclockwise, checking the accuracy of rotation on the zero mark of the lubber ring at each position. As a result of this test, the difference between the transmitted bearing and the indicated bearing shall not exceed the requirements specified in 3.16.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts. -

5.1.1 Plotting boards. -

5.1.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical and mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations applicable to the mode of transportation.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the

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contractor's commercial practice. The information shall include nomenclature, contract or order number, contractor's name, and destination.

5.1.2 Onboard repair parts. - The onboard repair parts shall be preserved and packaged level A; packed level C, and marked levels A and C, respectively, in accordance with Specification MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation, packaging, packing, and marking for shipment shall be specified by the procuring activity (see 6.2).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment which may be required when procurement is made by a Government activity (see 6.2):

5.2.1.1 Preservation, packaging, packing, and marking. - The equipment and accessories, repair parts, and technical publications shall be preserved and packaged level A or C, packed level A or B as specified (see 6.2), and marked in accordance with Specification MIL-E-17555.)

6. NOTES

6.1 Intended use - Summary plotting boards covered by this specification are intended for installation in combat information centers and similar spaces in Naval ships. The plotting board is used for providing a means for recording data and plotting points and courses either in terms of true bearing, or by reference to the own ship's course indicator, in terms of relative bearing.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity of equipment required.
(c) Quantity of technical manuals required
(d) Supply demand control point for repair parts (see 3.2).
(e) Type of synchro motor required (see 3.7).
(f) Type of plotting top required (see 3.15).
(g) Preservation, packaging, packing, and marking requirements if other than specified in 5 1 (see 5.2)

Notice - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Ships
(Project 6605-N037Sh)

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SPECIFICATION ANALYSIS SHEET

Instructions

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

Specification

Organization	City	State

Contract No.

Quantity of Items Procured	Dollar Amount $

Material procured under a direct Government contract ☐ or a subcontract ☐

1. Has any part of the specification created problems or required interpretation in procurement?
   a. Give paragraph number and wording.

   b. Recommendations for correcting the deficiencies.

2. Comment on any specification requirement considered too rigid.

3. Is the specification restrictive? ☐ Yes ☐ No If the answer is "Yes", in what way?

4. Remarks (Attach any pertinent data which may be of use in improving this specification.)
Place this form and papers in an envelope and send to the Bureau.

Submitted by (Print name and activity)	Date

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